

Highlights

Two bases for a voxel, the medium and the wavefield: the error of elastic volume-integral scattering, isolated on a space-filling layer of cubes

T. M. Nestor

- A voxel needs two bases: one for the medium and one for the wavefield
- On a layer of cubes the discretisation error is isolated, with no shape error
- The order is $\min(2p+2, 2r+2)$ for field degree p and medium degree r
- The single-site T-matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

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T. M. Nestor

Independent researcher, Australia

Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral (coupled-dipole) methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves, which cannot be measured where no exact solution exists. We answer on a layer of voxels: because cubes tile space, n planes of voxels are the layer itself, with no shape error, and its exact solution isolates the discretisation error. The source-averaged coupling equals the continuum's exactly; the cube's single-site T -matrix follows from distributional moments of the Green's tensor. With the wavefield expanded to Legendre degree p and the medium's contrast to degree r in each voxel, the scheme converges at order $\min(2p + 2, 2r + 2)$, with leading error $c_p(kd)^{2p+2}$ in closed form: $c_0 = 1/12$, $c_1 = 1/720$, $c_2 = 1/100800$; the orders hold for SV and SH waves and at oblique incidence, to $p = 2$. The single-site T -matrix's second-order error equals the relative projection error of the medium at degree $\min(p, r)$, exactly in the long-wave limit: the two bases must be matched,

and the error is known from the medium before anything is solved. Where the medium is piecewise constant on the voxels it vanishes: in a randomly stratified layer one first-moment voxel per layer reaches 10^{-6} . A smoothly graded sphere converges at second order against its exact solution. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background, which the model prescribes. The other is the *wavefield*: the displacement and

17 strain inside the voxel, which the calculation must find. Each is expanded
 18 in a basis on the cell, and the two bases are separate choices. Here both
 19 are Legendre polynomials, the wavefield to degree p and the contrast to
 20 degree r ; the usual scheme, a uniform field in a cell of constant properties,
 21 is $p = r = 0$. Neither choice is new in itself. In higher-order volume integral
 22 methods for electromagnetics the material within an element is represented
 23 by polynomials whose orders are independent of those of the field [1], as it
 24 is in finite elements, and voxels that carry a linear field are in use [2]. What
 25 has been missing is the law that relates the two bases: the order a given pair
 26 attains, and the error it leaves. This paper asks which basis each of the two
 27 needs, and what error a given pair (p, r) leaves. The answer decides the cost
 28 of everything built on it, because a voxel that represents its contents more
 29 accurately can be larger, and the number of voxels, not the size of each one's
 30 state, sets the cost of the multiple-scattering solve.

31 *Why the question is hard to answer.* For the media such a scheme is built for,
 32 there is no exact solution. Its error cannot be measured against the answer.
 33 Comparison with another numerical method mixes the errors of both. And a
 34 refinement study can mislead, because a voxel scheme can converge smoothly
 35 to the wrong limit. §4 shows how that happens with the most natural choice
 36 of coupling, the Green's tensor evaluated between voxel centres: it carries
 37 a local term whose size per unit volume does not shrink as the voxels do,
 38 and unless the single voxel's response is matched to cancel it, refinement
 39 converges to the wrong answer without revealing it.

40 *The method: a layer of voxels.* This paper therefore answers the question
 41 in the simplest configuration in which a voxel scheme runs in full (every

voxel, every coupling, the whole multiple-scattering series, at normal and
 at oblique incidence) and the exact answer is nevertheless known: a layer
 filled with voxels. Because cubes tile space, a layer of identical voxels does
 not approximate a homogeneous layer; it *is* one. Any difference between
 its reflection and transmission and those of the continuous layer is therefore
 error of the discretisation alone, with no modelling error and no shape error
 behind which to hide. A sphere, the usual test object for volume-integral and
 discrete-dipole methods [3], cannot give this: its error mixes the discretisation
 with the staircase approximation of a curved surface by cubes. The layer
 plays the part of the patch test in finite elements: reproducing a uniform
 medium correctly is necessary, though not sufficient, for a scheme to be
 trusted in a heterogeneous one. The experiment is then made harder in
 three steps, each with an exact solution of its own: a randomly stratified
 layer, in which every cell is homogeneous but the cells differ; a layer whose
 properties vary smoothly within every cell, which is where the basis for the
 medium is tested; and a sphere whose contrast falls smoothly to zero at its
 surface, which carries the layer's prediction into three dimensions without a
 staircase. What the layer establishes about the scheme's two ingredients, the
 coupling between voxels and the response of a single voxel, are properties of
 the ingredients, which are the same in any medium. A laterally heterogeneous
 medium remains to be tested.

The answer. (1) *The order is set by the weaker basis.* The scheme converges
 at order $\min(2p + 2, 2r + 2)$ in the cell size (§9). A contrast held constant in
 each cell therefore caps every voxel at second order in a medium that varies
 within the cells, however rich the basis of its field; and a contrast of higher

67 degree than the field is wasted. (2) *The leading error is in closed form, for*
 68 *every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$
 69 with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§8); the orders 2, 4 and 6 of
 70 $p = 0, 1$ and 2 hold for P, SV and SH waves at normal and oblique incidence.
 71 For the collocation scheme, the uniform-field voxel in its simplest form, the
 72 error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission (§7), and the
 73 opposite signs explain why no correction of the single site can improve both
 74 (§11). (3) *The error that remains is the single site's, and it is the projection*
 75 *error of the medium.* The single-site T -matrix sums the rescattering within a
 76 voxel to all orders, and the coupling between voxels is exact, so the multiple
 77 scattering itself is not approximated. But the T -matrix is that of the voxel as
 78 represented: its contrast projected to degree r and its field held to degree p .
 79 When the scattered field is separated by its order in the contrast, the error
 80 of the second-order term, in which the field scattered once within a voxel
 81 is scattered there again, equals the *relative projection error* of the medium:
 82 the fraction of the squared norm of its profile that polynomials of degree
 83 $\min(p, r)$ on the cells cannot represent (Eq. (38)). The equality is exact in
 84 the long-wave limit, where it is derived, and holds to within 1.3% for all nine
 85 pairs with $p, r \leq 2$ in the experiments (§9). Splitting that term confirms
 86 whose error it is: the part in which both scatterings lie in one voxel carries
 87 the error, and the part between voxels at most 0.2% of it. It shows what
 88 the wavefield basis is for: it must carry what the medium imprints on the
 89 field, not only the wave, which at these frequencies varies little across a cell.
 90 (4) *The two bases must be matched, and the error is known in advance.* The
 91 relative projection error is a property of the medium and the grid alone, so

92 the error of a calculation is known before anything is solved, and the cell size
 93 and the degree that a required accuracy needs can be chosen beforehand.
 94 (5) *Where the medium is piecewise constant on the cells the projection error*
 95 *vanishes*, a constant suffices for the medium, and the degree of the field
 96 alone sets the order: in a randomly stratified model one first-moment voxel
 97 per model layer reaches a reflection accuracy of 10^{-6} , where the uniform-
 98 field voxel needs about forty (§9), and in the uniform layer it allows voxels
 99 58 times larger in each direction (§11). (6) *In three dimensions* the smoothly
 100 graded sphere converges at second order with the mean-only voxel, as the
 101 layer predicts (§10), and at fourth order with the first-moment voxel in the
 102 companion paper [4].

103 *What makes the experiment exact.* The answer above can be trusted only if
 104 nothing but the two bases is approximated. Three results, set against the
 105 coupled-dipole method described below, secure that. (a) *An exact coupling.*
 106 The integrated Green tensor and the filtered coupled dipoles approximate the
 107 field of a cell; for cubes the source-cell average is exact, because the cube's
 108 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and
 109 at normal incidence the lattice kernel is in closed form with no lattice sum.
 110 (b) *No polarisability to choose.* The polarisability, a modelling choice in the
 111 coupled-dipole method, is here a derived object (§6): the moments of the
 112 Green's tensor over the cube, taken as distributions, yield Eshelby's δ term
 113 without adding it by hand, the cube's departure from the sphere as a pure
 114 cubic anisotropy, and the closure's error against exact scattering by a sphere
 115 as a real $(ka)^2$ term in closed form. The self term the single site carries is
 116 the one the averaged kernel omits, so the discrete system is a collocation of

117 the continuum equation. (c) *The elastic state.* A voxel carries a force and
118 a moment, so its state has nine components, the coupling is a 9×9 tensor,
119 and its local term is the depolarisation of a plate.

120 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
121 *space*, so a uniform region discretised into cubes is still that uniform region,
122 whereas spheres or point scatterers can only approximate it and carry a pack-
123 ing error of their own. Their *regular lattice* makes the interaction between
124 cells translation invariant, so the coupling is a convolution (Toeplitz between
125 planes, Bloch-periodic within them), and the whole system is solved by fast
126 transforms. And a cube small compared with the wavelength has a *single-site*
127 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
128 approximation a voxel introduces is the internal field it assumes.

129 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of
130 small scatterers that interact through the background Green’s function is the
131 multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics
132 it became a numerical method as the coupled-dipole method, or discrete
133 dipole approximation (DDA): each cell of a cubic grid is a polarisable point
134 dipole driven by the incident field and by the fields of every other dipole
135 [7, 8]. The method is a collocation of the volume integral equation for the
136 electric field, with the cell’s self-interaction absorbed into its polarisability
137 [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns
138 on two ingredients, and this paper takes up both for elasticity. The first is the
139 polarisability. The Clausius–Mossotti value is the static response of a sphere;
140 the lattice dispersion relation of Draine and Goodman [11] adjusts it so that
141 an infinite lattice propagates waves as the continuum does to a given order

142 in kd ; Rahmani et al. [12] instead derive it from the exact static solution for
 143 the whole body, so that the method is exact in the long-wavelength limit.
 144 The second is the coupling. Integrating the Green tensor over the source
 145 cell rather than sampling it at the centre improves the accuracy, most of
 146 all for strong contrasts [13], and the filtered coupled dipoles of Piller and
 147 Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace
 148 the point dipole by a source of finite extent whose spectrum is band-limited
 149 to the grid. The coupling summed over a lattice extends the method to
 150 periodic structures [16]. Its error converges at second order in the dipole
 151 spacing when shape errors are absent [3], and for a cube, the one shape
 152 a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach
 153 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. On the
 154 mathematical side, Costabel et al. [18] show that the method is a finite section
 155 of an infinite block Toeplitz matrix, stable for a large range of the contrast
 156 but not for all of it, and Costabel [19] that in a one-dimensional model it
 157 converges at second order in the interior, with a boundary layer in which
 158 the consistency error does not vanish. The layer studied here is the elastic
 159 counterpart of that cube: a body the grid represents exactly, so that the
 160 discretisation error can be isolated. What this paper adds is the closed form:
 161 for cells that tile space, the coupling that makes the lattice reproduce the
 162 continuum exactly, and the order and the constant of the error that remains,
 163 as functions of the two bases. In the terms used here the coupled-dipole
 164 method is the pair $p = r = 0$, and its refinements improve the polarisability
 165 and the coupling while leaving both bases constant in the cell.

166 *Its elastic counterparts.* The elastic problem has the same volume integral
 167 equation [20], and every cell-based solver of it is a coupled-dipole method
 168 whether or not it carries the name. Nestor [21] divided a two-and-a-half-
 169 dimensional medium into cells that interact through the background Green's
 170 tensor, each carrying a contrast renormalised by its own near field, with
 171 the multiple-scattering series summed. Kanaun and Levin [22] discretise the
 172 inclusion problem with Gaussian approximating functions whose matrix ele-
 173 ments are analytic. Yang et al. [23] solve a stress-velocity form of the integral
 174 equation on a grid by conjugate gradients with fast Fourier transforms, and
 175 Touhei [24] does the same for a half space with a generalised Fourier trans-
 176 form. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and
 177 inverse problems on the transition operator of the discretised Lippmann-
 178 Schwinger equation, in the anisotropic elastic case with a nine-component
 179 state vector. The elastic problem adds a structure the scalar and electro-
 180 magnetic ones lack. A cubic voxel responds to the incident field with a force
 181 (through its density contrast) and a moment (through its stiffness contrast),
 182 so the state at a voxel has nine components (displacement and strain), and
 183 the single site's self-interaction is the elastic depolarisation of a cube, the
 184 discrete counterpart of the inclusion tensor of Eshelby [27].

185 *This paper.* It extends the formulation of Nestor [21] to cubic voxels, and
 186 is ordered as the experiment is built. §2 sets out the discrete medium and
 187 §3 the continuous layer it must reproduce. §4 shows why the point coupling
 188 fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives
 189 the single-site T -matrix from moment integrals, tests it against the exact
 190 sphere, and shows that with it the discrete system is a collocation of the

191 continuum equation. The wavefield basis comes next: §7 gives the second-
 192 order error of the uniform field in closed form, and §8 the voxel whose field
 193 carries Legendre moments to degree p , with its order $2p+2$ and its constant,
 194 for incident P and S at normal and oblique incidence. The medium basis
 195 follows in §9: a randomly stratified layer, for which a constant in each cell
 196 is exact, and a medium that varies within a cell, for which the contrast is
 197 expanded to degree r , the order becomes $\min(2p+2, 2r+2)$ and the error of
 198 the single-site T -matrix is the relative projection error of the medium. §10
 199 carries the prediction to a sphere, §11 discusses what the results imply for
 200 voxel schemes in general, and §12 closes with why an error known in advance
 201 is worth having. Every result is derived symbolically in Mathematica and
 202 checked against an independent Python implementation or an exact solution.

203 2. Setting

204 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 205 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 206 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 207 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 208 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 209 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

210 displacement followed by engineering strain. A voxel responds with a force
 211 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 212 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source

213 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

214 with ψ^0 the incident state.

215 *Where the kernel comes from.* Write the medium as the background plus
 216 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 217 $\rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Split-
 218 ting $\rho = \rho_0 + \Delta \rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta \lambda \delta_{bc} \delta_{de} + \Delta \mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$
 219 the contrast stiffness, and moving the contrast terms to the right leaves the
 220 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta \rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

221 a force from the density contrast, and the divergence of the contrast stress m ,
 222 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the
 223 background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term
 224 integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

225 the boundary term vanishing because m is confined to the scatterers, and the
 226 last step holding because $\mathbf{G}$ depends only on the separation. The moment
 227 therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when
 228 the moment is the contrast stress itself. With the moment tensor defined
 229 instead through its equivalent body force $-\partial_c [M_{bc} \delta(\mathbf{x} - \mathbf{y})]$, as in Aki and
 230 Richards [28], the same field reads $M_{bc} \partial G_{ab} / \partial y_c$, the source derivative, with

231 $M = -m$. The two are one convention written two ways; what matters is that
 232 the derivative in the kernel and the sign in the contrast operator are chosen
 233 together. The nine-component kernel follows (Fig. 1): its force columns are
 234 $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over
 235 (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are
 236 those entries and its strain rows their receiver derivatives, symmetrised in
 237 the same way and doubled on the shear rows to give engineering strain. In
 238 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

239 where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$
 240 and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

241 *Averaging, and collocation.* Three operations in this paper are averages over
 242 a cell, and they are distinct. The *source-cell average* integrates the kernel
 243 over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ is the field at $\mathbf{r}$
 244 of a uniform polarisation of the source cell. Every scheme in this paper uses
 245 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 246 is imposed: at the receiving cell's centre, with the internal field of each cell
 247 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 248 *average* imposes the equation in the mean over the receiving cell instead;
 249 with a uniform internal field that is the Galerkin scheme of degree zero,
 250 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 251 term that refers to the internal field and not to the test, and they differ only
 252 in the test function (a delta at the centre against the cell's indicator) and
 253 in the constant of their $O((kd)^2)$ error (Table 3). The *cell-mean contrast*,

254 finally, is an average of the material, not of the field: where the medium
 255 varies within a cell, the contrast is projected onto the cell's basis, its mean
 256 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 257 of the coupled-dipole method, the point dipole is collocation with the point
 258 kernel, the integrated Green tensor adds the source-cell average, and the
 259 filtered coupled dipoles replace the point source by one of finite extent.

260 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 261 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 262 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 263 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 264 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

265 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 266 join the centres of the voxels in one plane (a square lattice of pitch d), and
 267 m is the separation of the two planes in units of d (Fig. 2a,b). The question
 268 is whether, and how fast, this chain converges to the continuous layer as
 269 $n \rightarrow \infty$ at fixed D .

270 *Implementations.* Every result is derived symbolically in Mathematica and
 271 compared with an independent numerical implementation or an exact solu-
 272 tion. “The package” below is the Python implementation of the voxel scheme
 273 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 274 propagator, which serve both as the object under test and, where validated,
 275 as a second implementation.

276 3. The continuous layer and its thin-layer expansion

277 At normal incidence a plane force drives a one-dimensional problem per
 278 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 279 Green's function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer's reflection and transmission
 280 follow from continuity of displacement and traction at its faces; they agree
 281 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 282 P and S (`ContinuumLimit_Reference.wl`).

283 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 284 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 285 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coeffi-
 286 cient expands as

$$R = \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (7)$$

287 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 288 coefficient symbolically and the truncated series against the exact layer).
 289 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

290 to first order a thin layer is an interface across which the traction jumps by
 291 the excess inertia and the displacement by the excess compliance. At oblique
 292 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 293 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 294 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients

295 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 296 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 297 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 298 a single plane of voxels must reproduce order by order. A plane of voxels
 299 that carry Legendre moments of their internal field to degree p reproduces
 300 Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 301 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 302 first-moment voxel all four terms shown.

303 4. The specular sums of the point Green's tensor

304 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 305 of every voxel of a plane m planes away. In that form it is a poor object to
 306 study. The Green's tensor decays slowly with distance, so the sum converges
 307 slowly, and it hides the one question that matters here: which part of it is
 308 the continuum, and which part is the discretisation. The Poisson summation
 309 formula [29] answers both at once, and is the standard tool for lattice sums
 310 of wave fields [30]. For a function f on the plane and the square lattice of
 311 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

312 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 313 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3).
 314 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

315 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 316 between planes a distance z apart. Each term has a physical meaning. Trans-
 317 formed laterally, the field of a plane of sources is a superposition of plane
 318 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 319 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 320 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 321 far beyond the wavenumber of the field when the voxels are small compared
 322 with the wavelength, so it is evanescent and decays away from the plane as
 323 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 324 the lattice sum exactly into the continuum and a discretisation correction,
 325 and the correction converges after a few orders.

326 *The local term that survives.* In the strain-from-moment block the evanescent
 327 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 328 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 329 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 330 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 331 planes in slab order, the static lattice sum of the point kernel is exactly
 332 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 333 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 334 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 335 reproduces the continuum only if that remainder is cancelled by the voxel's
 336 own self term.

337 5. The tiling identity

338 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 339 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 340 Each Poisson order in Eq. (10) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

341 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 342 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 343 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 4).
 344 Two exact statements follow.

345 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

346 the continuum's plane-wave term times the vertical form factor. The pack-
 347 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 348 `AveragedSums.wl`).

349 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 350 is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

351 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 352 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 353 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 354 three normal strain-from-moment entries, whose local weights $-1/M_P$ and

355 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 356 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 357 with every cell integral evaluated analytically (below).

358 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 359 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

360 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 361 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 362 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 363 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 364 and the quadrature order are raised; Eq. (14) is now the package’s kernel at
 365 normal incidence.

366 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 367 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 368 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 369 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 370 which never contain the origin, and there the two-dimensional antiderivative
 371 in the face coordinates is continuous, so the face integral is a sum over the
 372 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 373 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 374 *not* usable for repeated derivatives: its arctangent terms jump across the
 375 coordinate planes through the origin, and differentiating twice there discards
 376 exactly that term.) The self term so obtained reproduces the classical closed
 377 form of the cube’s depolarisation to machine precision.

378 6. The single site: the cube T -matrix from moment integrals

379 Every error of a voxel scheme is made at the single site (§11), so the single
380 site is derived here rather than imported. The classical route imports it. For
381 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
382 uniform load uniform, with a depolarisation tensor in closed form [27, 31, 32];
383 its dynamic generalisation is known for spheres and ellipsoids [33, 34]. For
384 a cube or a general polyhedron the static field of a uniform eigenstrain is
385 also in closed form [35, 36, 37, 38], but the strain it produces is not uniform,
386 so there is no exact uniform-strain inclusion to import. In scattering theory
387 the same object appears as the self-consistent or effective-field closure of
388 the volume integral equation [39, 20, 40]: a first-order (Born) scatterer [41]
389 renormalised by the geometric series of its own near field, the step that turns
390 a contrast into an effective contrast. Nestor [21] introduced the single site
391 in this form, summing the series of the cell’s self integral, with the singular
392 near field evaluated over an infinitesimal cell. The moments below evaluate
393 the same self integral over the finite cube, as a distribution. The elastic T -
394 matrix of Waterman [42] represents a scatterer in spherical multipoles, the
395 natural basis for a sphere; a voxel in a lattice needs one compatible with the
396 lattice, and the moments over the cube provide it. The construction below
397 makes the approximation of a voxel explicit rather than implicit: it derives
398 the cube’s T -matrix from the moments of the elastodynamic Green’s tensor
399 over the cube, and identifies the uniform-strain closure as one truncation of
400 an exact hierarchy.

401 *6.1. The closed hierarchy*

402 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 403 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 404 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 405 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed
 406 system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (16)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

407 all fields evaluated at the centre. Its coefficients are six moments of the
 408 background Green’s tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (18)$$

409 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 410 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
 411 it, so it is fixed by convention. The moments are graded by (D, W) , the
 412 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
 413 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

414 *6.2. The moments as distributions*

415 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (19)$$

416 [43, 28]. Expanding g_c in powers of r reduces every moment in Eq. (18) to a
417 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (20)$$

418 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
419 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
420 gral depends on the shape of the region excluded around the origin. This is
421 the source of the δ -function term of the inclusion problem, which a numerical
422 quadrature misses. Without it the depolarisation is wrong; for a sphere it
423 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
424 are distributional from the start, and each is moved onto the voxel's faces by
425 pairing with the test function $w\mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (21)$$

426 The surface term is evaluated on the faces, a distance h from the origin, where
427 every derivative of r^m is smooth. The recursion lowers D and W together
428 and ends on convergent integrals. The face integrals are done in closed form
429 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (22)$$

430 which the face route satisfies exactly and a spherical excision returns as
431 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell

432 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 433 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 434 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 435 Because every power of r is paired this way, the dynamic moments keep the
 436 even-power terms of the expansion (k^2r , r^3 , ...) as well as the odd ones.
 437 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 438 those terms, at a real $O((kh)^2)$ cost (§6.4).

439 6.3. Parity, and the nine-component block

440 The cube is centrosymmetric. A moment that couples a field quantity of
 441 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 442 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (16) contains the first gradient alone: ∂u is
 443 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 444 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 445 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 446 by Eq. (16), and *no treatment of the second-gradient block (17) can change*
 447 *it*. What Eq. (17) contributes reaches the far field only through the $O((kh)^2)$
 448 coupling in Eq. (15). That is the order, and the origin (the variation of the
 449 field across the voxel), of the collocation error of §7, which the first-moment
 450 voxel of §8 removes.

451 With $\Delta\rho = 0$, Eq. (16) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 452 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 453 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are

454 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (23)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (24)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (25)$$

455 The density channel is the displacement block of Eq. (15) with the second
 456 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 457 of $\mathbf{G}$.

458 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 459 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 460 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 461 latational channel, Eq. (23), is the sphere's exactly: its eigenvalue involves
 462 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 463 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 464 the shape. The cube's shear channels differ from each other. Their isotropic
 465 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (26)$$

466 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 467 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 468 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 469 and the diagonal shear is the more depolarised. At the background of §7

470 $(\nu = 7/32)$, $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 471 0.4960.

472 *Where Eshelby enters.* Equations (23)–(25) follow from the moment integrals
 473 and the group theory alone. No ellipsoidal result is used, and no δ term is
 474 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 475 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 476 the uniformity theorem that ansatz is exact if and only if the inclusion is
 477 an ellipsoid. For a sphere the construction is therefore exact in the static
 478 limit, and for a cube it is an approximation. The hierarchy (15)–(17) states
 479 that approximation as the omitted blocks, rather than as an unquantified
 480 shape correction. A polynomial expansion about the centre cannot represent
 481 the edge and corner concentrations of an isolated cube's internal field at any
 482 finite degree. A voxel in a uniform layer has no such concentrations: its
 483 neighbours have the same properties, so its internal field is the continuum's,
 484 which is smooth. That is why a low-degree internal field suffices in a lattice
 485 (§§7–8) although it is only an approximation for an isolated cube.

486 *Against the exact sphere.* These statements are tested against the exact elas-
 487 tic Mie solution for a sphere, with no Eshelby formula in between. The
 488 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 489 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 490 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 491 route of Eq. (21) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 492 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 493 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 494 isotropic shear average, Eq. (26), is the depolarisation read from Mie's E_2

495 to 10^{-8} . The same moment taken as a principal value, over the ball with
 496 a small sphere excluded around the origin, has zero depolarisation in every
 497 channel. The whole static response of the sphere is the δ term.

498 At finite frequency the closure uses the dynamic ball moments, which are
 499 in closed form. It departs from Mie at second order in ka and first order in
 500 the contrast, and the departure is closed-form too: the ball's moments are
 501 elementary, and Mie's low-frequency series with the contrast kept exact gives
 502 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (27)$$

503 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 504 the radiation damping, agree exactly in all three channels. So the closure's
 505 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 506 own. That is the order the hierarchy assigns to the omitted second gradi-
 507 ents (§6.3), and it has the same origin, the variation of the field across the
 508 scatterer, as the collocation error of §7. Eq. (27) is derived symbolically
 509 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 510 the numerical Mie solve and an independent implementation of the closure
 511 to 5×10^{-4} , which is the size of the next order in the contrast.

512 6.4. The collocation closure

513 The collocation closure takes the internal field of a voxel as uniform, the
 514 truncation of §6.3 for the density and stiffness channels together. The voxel's

own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (28)$$

With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (29)$$

an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
exact cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
 so Eq. (29) is precisely the collocation of the continuum Lippmann–Schwinger
 equation with a strain piecewise constant over voxels. It does not contain
 $\mathbf{S}_{\text{self}}$ at all.

The package’s cube T -matrix is this closure statically: both blocks agree
 with Eq. (28) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 through its far-field form-factor corrections. Its self integrals are exact in the
 static and radiation parts but omit the real even-power terms of the Green’s
 tensor’s expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 self term, whose series agrees with the closed-form elastodynamic tensor to
 10^{-14} . By Eq. (29) that error cannot enter the closure scheme.

533 7. Convergence, and the closed-form error

534 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
535 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
536 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
537 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
538 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (29) and with the
539 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
540 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
541 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
542 the exact solution and n the refinement; the orders in the legend of Fig. 5
543 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
544 with their standard errors. The two agree where a sequence is asymptotic
545 and differ where it is not, which the apparent orders show.

546 *The leading error in closed form.* In Eq. (29) the kernel is integrated exactly;
547 the only approximation is the internal field's variation across a voxel. At first
548 order in the contrast the discrete scattered field is therefore a midpoint rule
549 applied to the continuum's Born integral, with the kernel's cell integral exact.
550 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
551 it is flat. With k the background wavenumber of the wave concerned (k_P for
552 an incident P wave, k_S for an incident S wave), R the reflection coefficient
553 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4),$$

(30)

554 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
555 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
556 and those of Eq. (33) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
557 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
558 the transmission constant grows to -0.092 .

559 *Why a uniform-field single site stops at second order.* The two constants have
560 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
561 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
562 reduce the reflection error by a factor 4.45 and *increase* the transmission
563 error by 1.7, at every n . The missing term in Eq. (30) is the first moment
564 of the internal field over each voxel, which radiates through the derivative of
565 the kernel. A fourth-order scheme must carry it: a single site with a gradient
566 channel, and a kernel whose cell integrals include the kernel's derivative.

567 **8. Fourth order and beyond: Legendre moments of the voxel field**

568 Let each voxel carry the mean *and* the first moment of its internal field.
569 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) =$
570 $P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w =$
571 $\sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the
572 polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the
573 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (31)$$

574 with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz\right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (32)$$

575 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 576 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$
 577 and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form.
 578 Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 con-
 579 verges at fourth order in both reflection and transmission (`ContinuumLimit_`
`FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (32). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

580

581 the 2 m layer 6500 times more accurately than a single plane of uniform ones.
 582 The collocation column is the scheme of §6 and reproduces the chain of the
 583 full 9×9 system to all printed digits.

584 *Higher moments.* Carrying the Legendre moments to degree p changes noth-
 585 ing in the construction: the moments and the same-cell double integrals of
 586 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-

587 degree solver reproduces the chain above exactly. The second-moment voxel
 588 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
 589 eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment_`
 590 `w1`); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

591

592 *The leading error in closed form.* Without scattering, the Galerkin solution
 593 is the L^2 projection Π of the incident field onto each voxel's basis, and the
 594 kernel is integrated over each voxel exactly; at first order in the contrast the
 595 error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field
 596 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
 597 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and
 598 every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] \, ds}{\int_{-h}^h e^{i(q+k)s} \, ds}, \quad (33)$$

599 with Π the centre value for collocation, the cell mean for the mean-only voxel
 600 and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_`

601 `FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results
are in Table 3. The chain of Eq. (32) reproduces the forms for collocation

Table 3: The leading error of the discrete layer in closed form, Eq. (33): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

602
603 and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being
604 the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7}
605 at a contrast reduced by 10^{-6} . The constant for degree p follows from the
606 Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops
607 the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment
608 against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n/(2n+1)!!$ and
609 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (34)$$

610 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
611 mission for odd p , opposite signs for even p . The first-moment voxel's leading
612 error therefore does not depend on the direction in which the field is observed.

Eq. (34) is checked symbolically for $p = 0$ to 5 (every lower power of kd vanishes) in Mathematica and, independently, in sympy.

One plane of voxels and the thin-layer series. Refining at fixed thickness is one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$, $d = D$) the series in D of the reflection coefficient, computed exactly at the full contrast, coincides with that of the exact layer, Eq. (7), term by term through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.w1`). Each further Legendre moment adds two terms of the thin-layer series.

In three dimensions. The three-dimensional first-moment voxel carries the nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns in all, tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (32) (`ContinuumLimit_Reduction3D.w1`). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and $x/h, y/h$ are odd there. The mean and the z -moment carry weights that are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in z alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift,

637 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h / p\pi + O(k_x^2 h^2)$, so the tiling identity holds
638 only to $O(k_{\parallel} h)$.

639 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
640 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (35)$$

641 with L_a the lateral Legendre factors. The whole-space tensor decays as
642 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each
643 entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma +$
644 $\mathrm{i}\beta \text{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double
645 averaging makes the lattice sum converge without Ewald splitting, to three
646 digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20°
and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (35). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

647
648 second order and the first-moment voxel fourth order in both the reflected

649 P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$
650 and the evanescent coupling of the gradient channels do not cost an order.
651 At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (32) to all printed digits.

652 *Incident S.* Nothing in the construction is specific to P. For a normally in-
653 cident S wave the layer is the one-dimensional problem of Eq. (32) with
654 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes
655 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
656 transmission, and the full 9×9 construction with an incident SH wave re-
657 produces that one-dimensional result to all printed digits. At oblique inci-
658 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

659

660 of Eqs. (30) and (33), are those of the incident P wave.

661 *Degree two at oblique incidence.* In three dimensions the voxel of degree 2
662 carries, on each of the nine components of its state, the ten Legendre products
663 of total degree at most two in (z, x, y) . In the construction of Eq. (35) it
664 converges at sixth order against the exact layer for incident P, SV and SH

665 waves (Table 6; `scripts/measure_oblique_degree.py`). The test is run at
666 $\omega = 3000 \text{ rad s}^{-1}$ ($k_P D = 1.2$, $k_S D = 2$), ten times the frequency of the other
667 tables, because at 300 rad s^{-1} a sixth-order error on these grids lies below
668 the round-off of double precision. The exact reflection coefficient is that of
669 Kennett’s recursion, an independent algorithm, and the same computation
670 with the bases of degree 0 and 1 returns orders 2 and 4 in all five cases.
671 At normal incidence the error on four planes, 6.95×10^{-9} , is the $c_2(kd)^6 =$
672 7.2×10^{-9} of Eq. (34) to 4%. Doubling the number of lattice orders retained
673 in Eq. (35) changes the errors by less than 1%. Unlike the degrees 0 and 1,
674 this degree is implemented once at oblique incidence; its check is the exact
solution.

Table 6: The voxel of degree 2 (ten Legendre products per state component) at $\omega = 3000 \text{ rad s}^{-1}$: relative error of the same-type reflection coefficient against the exact layer on n planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

incident wave	error, $p = 2$			order	orders
	$n = 1$	$n = 2$	$n = 4$	$p = 2$	$p = 0$ / $p = 1$
P, normal: R_{PP}	3.4×10^{-5}	4.6×10^{-7}	7.0×10^{-9}	6.05	2.04 / 4.05
P at 20° : R_{PP}	7.3×10^{-6}	1.0×10^{-7}	1.5×10^{-9}	6.04	2.04 / 4.06
SV at 20° : R_{SS}	2.2×10^{-4}	2.3×10^{-6}	3.4×10^{-8}	6.11	2.10 / 4.14
SH at 20° : R_{SH}	2.3×10^{-4}	2.5×10^{-6}	3.6×10^{-8}	6.11	2.10 / 4.14
SH at 40° : R_{SH}	3.0×10^{-4}	3.6×10^{-6}	5.2×10^{-8}	6.09	2.12 / 4.05

675

676 The spectral construction is implemented twice. The second implementa-
677 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
678 nition of the scheme: it takes the depth dependence of the transformed kernel

679 from an independent residue construction, obtains the local weight as a nu-
 680 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
 681 ble integral by quadrature and separates P from S by sampling the upgoing
 682 field at two depths. The raw reflected displacements of the two agree to $9 \times$
 683 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
 684 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
 685 scheme of any degree is also implemented twice. The second implementation
 686 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
 687 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
 688 from its own interface system, in double precision. Its errors agree with the
 689 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
 690 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 691 6.00; for one plane as the layer thins its differences from the exact layer fall
 692 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 693 Figure 5(a,b) shows the convergence, with the orders fitted to it.

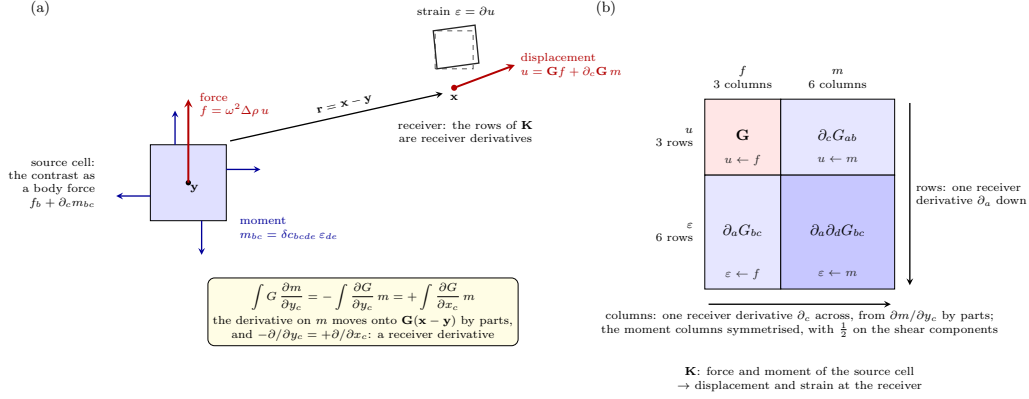


Figure 1: Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m , the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε . The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled u , three rows, and strain, six rows, and columns labelled f , three columns, and m , six columns; the blocks read $\mathbf{G}$, the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

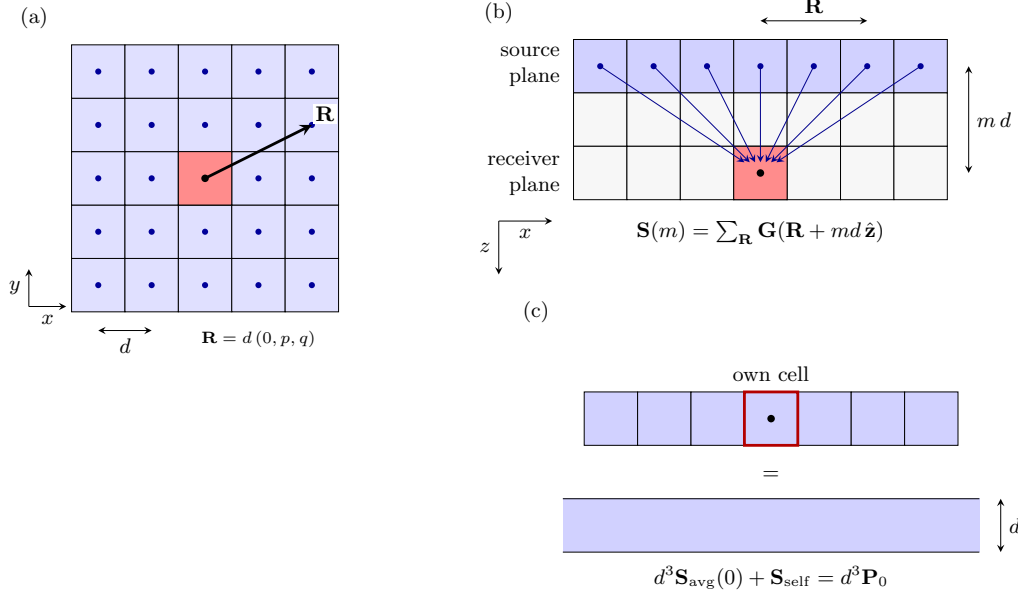


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (shaded darker) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

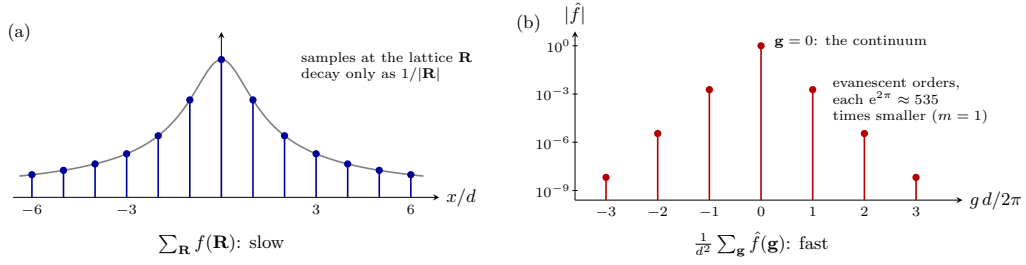


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

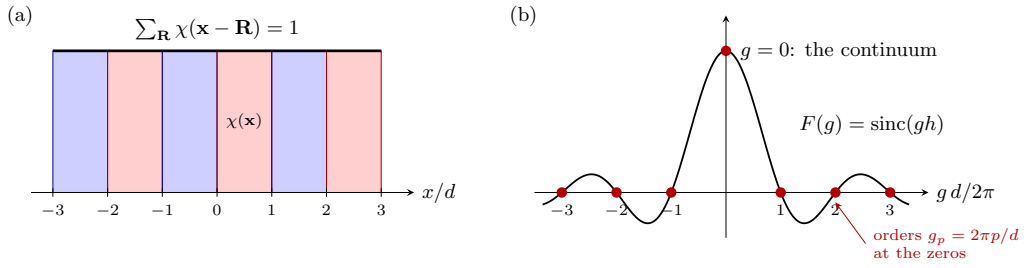


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

694 9. A heterogeneous layer

695 A heterogeneous medium is not a layer of identical cells. The simplest
696 heterogeneous case with an exact answer is a stratified model built from layers
697 of constant properties, with real interfaces between them: the form of many
698 layered crustal velocity models. The last test takes such a model. A layer
699 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
700 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
701 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
702 stratification with physical contrasts. The voxels refine the model's layering,
703 with m planes of voxels in each model layer, so no voxel straddles a model
704 interface: the natural discretisation when the model's interfaces are known.
705 At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
706 reverberates between the model interfaces. The exact answer is the product
707 of the model layers' propagators. Its same-type reflection coefficients agree
708 with those of the reflection–transmission recursion of Kennett [44], computed
709 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
710 now changing from plane to plane; nothing else changes. The errors are in
711 Table 7 and Fig. 5(c).

712 *The interfaces cost nothing.* Every order is the one measured for the uniform
713 layer. At a model interface the strain of the true field jumps. A voxel basis
714 that is discontinuous from voxel to voxel represents that jump exactly when
715 the interface lies on a voxel boundary, so the interface adds no error of its
716 own. The only requirement is that voxel boundaries coincide with the model's
717 interfaces. A voxel that straddled an interface would carry a contrast that is
718 not uniform within it, a case not examined here.

Table 7: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

719 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
720 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
721 layer. The first-moment voxel needs 8, one per model layer: the model's own
722 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
723 against 36), that is 3123 against 288 per vertical column of voxels, a factor
724 of eleven. In three dimensions, with the lateral grid refined as the vertical
725 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
726 unknowns by about 2×10^4 . The same model has been solved independently
727 in Python, with a contrast for each plane: the raw reflected displacements
728 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
729 at 20° , with and without the first moments. Lateral heterogeneity needs the
730 full three-dimensional solver; it is the first extension discussed below.

731 *A medium that varies within a cell.* The stratified model is exact on its own
732 grid: every cell is homogeneous. A smoothly varying medium is not, and
733 sampling it as cells of constant properties is an approximation of its own.
734 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
735 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
736 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
737 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (32)
738 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (36)$$

739 the same kernel moments, now to degree $p+r$, and nothing else new. Against
740 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
741 the order of convergence is

$$\min(2p+2, 2r+2), \quad (37)$$

742 or $2p+2$ when degree r represents the profile exactly (Table 8; `ContinuumLimit_`
743 `GradedContrast.wl`). A contrast that is constant in each cell therefore
744 caps every voxel at second order, however rich its field basis, and for the
745 smooth profile the voxel of degree 2 with a linear contrast has the error
746 of the voxel of degree 1 to within 0.06%. At first order in the contrast
747 this is what one expects: the contrast's projection error is orthogonal to
748 the polynomials of degree r , so against the smooth product of the inci-
749 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
750 the contrast must be represented to the degree of the field. The closed forms
751 of the construction are tested one by one by an independent implementa-

Table 8: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (37), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
bles the scheme without linearisation, evaluating the projected contrast at
the quadrature nodes, against its own 30-digit solution of the graded layer;
its errors agree with those of Table 8 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it
measures the orders of Eq. (37) again.

Which basis sets the error. Equation (37) gives the order. The size of the
error, and which of the two bases it belongs to, appear when the scattered
field is separated by its order in the contrast: the Born term T_1 , and the
second-order term T_2 in which the field scattered once is scattered again.
Both follow from the solution at contrasts scaled by $\pm\epsilon$ and $\pm 2\epsilon$, for the

765 scheme and for the exact layer alike. With $f(z)$ the profile and Π_q the L^2
 766 projection onto the Legendre polynomials of degree at most q in each cell,
 767 define the *relative projection error* of the medium at degree q ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (38)$$

768 the fraction of the squared norm of the profile that polynomials of degree q
 769 on the cells cannot represent. It is a property of the medium and the grid
 770 alone; it involves neither the wave nor the solution. For the smooth profile
 771 and every one of the nine pairs with $p, r \leq 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (39)$$

772 to within 1.3% at $\omega = 300 \text{ rad s}^{-1}$ on 4 to 16 cells: the ratio of the two sides
 773 is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2.
 774 The error of the Born term is smaller by two to six orders of magnitude. The
 775 error of the whole is therefore that of T_2 , the projection error times $|T_2|/|T_1|$
 776 (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7} , and Table 8
 777 has 1.4×10^{-7} .

778 Equation (39) is exact in the long-wave limit, for a reason that shows what
 779 each basis is for. The strain-strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$,
 780 is local as $kD \rightarrow 0$, and the incident strain ε_0 is then uniform across the layer,
 781 so the strain scattered once is $-(\Delta M/M_P) f(z) \varepsilon_0$: it is the profile itself. A
 782 cell holds the contrast as $\Pi_r f$ and the field to degree p , so it holds the once-
 783 scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order
 784 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) dz = \|\Pi_q f\|^2, \quad (40)$$

785 the last because $\Pi_q f$ lies in the range of Π_r . The relative error is $1 -$
 786 $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$. Because the kernel is local in this limit, the second
 787 scattering takes place in the cell of the first: T_2 is the second-order term of
 788 the single-site T -matrix, and Eq. (39) is a statement about the single site.
 789 The T -matrix sums the rescattering within a cell exactly for the cell as it
 790 is represented, a contrast $\Pi_r f$ and a field of degree p ; its error is that the
 791 cell is not the medium. The split is measured directly (`scripts/measure_`
 792 `layer_bases.py`): at $\omega = 300 \text{ rad s}^{-1}$ on 4, 8 and 16 cells, T_2 of the scheme
 793 and of the exact layer are each divided into the part whose two scatterings
 794 lie in one cell and the part between different cells. For all nine pairs the
 795 error of the same-cell part is 0.998 to 1.013 times $|T_2| \mathcal{E}_{\min(p,r)}$, and that of
 796 the part between cells at most 0.002 times it. The coupling between cells
 797 contributes only the departure from the limit. Measured, the ratio of the
 798 two sides of Eq. (39) for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2} ,
 799 4.3×10^{-3} , 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24, 0.12, 0.04$ and 0.012 : as
 800 $(kD)^2$ (`scripts/measure_layer_bases.py`). The pairing in Eq. (40) is ver-
 801 ified symbolically for all nine pairs, and the projection errors are computed
 802 exactly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each halv-
 803 ing of the cells.

804 Three consequences follow. The two bases must be matched: a contrast
 805 of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$
 806 have the errors of $(0, 0)$ to three digits), and a field of higher degree than
 807 the contrast improves only the Born term. The wavefield basis is there to
 808 carry what the medium imprints on the field, not only the wave: at these
 809 frequencies the wave itself varies little across a cell. And the error is known

810 from the medium, before anything is solved: $\mathcal{E}_q$ is a property of the profile
811 and the cells. It vanishes when the medium is piecewise constant on the cells,
812 which is the stratified model above; there the contrast needs no more than a
813 constant in each cell, and the degree of the field alone sets the order.

814 **10. A sphere: what the layer predicts in three dimensions**

815 The layer and the stratified layer are laterally uniform. A sphere is not,
816 and elastic scattering by a sphere, homogeneous [45, 46] or radially varying
817 [47], has an exact solution: it is the usual benchmark for volume-integral and
818 discrete-dipole methods. This section solves two spheres with the full three-
819 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
820 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
821 iterative solver, and a P wave is incident. For the graded sphere the voxel is
822 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
823 nine-component state, tested with the same constant, and the L^2 projection
824 of its contrast, which is its cell mean; every cell that overlaps the sphere is
825 kept, and the cells are coupled by cell-to-cell double integrals of the Green's
826 tensor, one block per orbit of the cube group [4]. For the sharp sphere
827 the voxel is the package's cube T -matrix, which differs from the collocation
828 closure only through its far-field corrections (§6.4) and has the same order,
829 each cell keeping the contrast of its centre. The far field is computed from
830 the solution for the incident plane wave itself, each voxel's displacement and
831 strain carrying the plane-wave phase at its centre; adding to that phase a
832 linear variation about the scatterer's centre, as a Taylor expansion of the
833 incident field would, counts the wave's variation twice and leaves an error of

834 order $(ka)^2$ that no refinement removes. The error is the largest difference
 835 between the scattered far field and the exact one over nine scattering angles
 836 from 11° to 169° , relative to the largest exact value, at a distance of 5×10^8
 837 radii for the graded sphere: at 5×10^4 radii the near-field terms that the
 838 asymptotic far field omits are 1.4×10^{-4} of the peak, 4% of the discretisation
 839 error at 12 voxels across and most of it at 24. The sharp sphere, whose errors
 840 are a hundred times larger, is observed at 5×10^4 radii.

841 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 842 its shape with every grid. A sphere whose contrast falls smoothly to zero at
 843 its surface does not: the voxels that cross the surface carry a vanishing con-
 844 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 845 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 846 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 847 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 848 the radial equations of Takeuchi and Saito [47], integrated across the shell
 849 from the core's regular solutions and matched to outgoing waves at $r = a$.
 850 The equations are derived symbolically from the equations of motion, and
 851 the solution is checked against a sphere of homogeneous concentric shells
 852 in closed form, by unitarity and reciprocity, and by an independent imple-
 853 mentation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_`
 854 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel
 855 carries the mean of the contrast over its cell. Table 9 shows the error falling
 856 at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (`scripts/pilot_`
 857 `graded_voxel_sphere.py`): the apparent order between successive grids is
 858 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders

859 of Fig. 5(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across the
860 error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
861 graded shell and the order is not yet asymptotic. This is the order the layer
862 predicts for the mean-only voxel, and the first test of that prediction on a
863 body with genuine three-dimensional structure.

864 What makes the sequence this clean is that the voxel carries a single error.
865 The collocation voxel of the sharp sphere applied to this body, each cell at
866 the profile's value at its centre and only the cells whose centres lie inside the
867 sphere kept, carries two: its own internal-field error and the error of sampling
868 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
869 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
870 2.0, 3.5 and 3.6 at $k_S a = 0.5$, and at $k_S a = 1$ the apparent orders are 1.7, 2.5
871 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
872 but its apparent order is not that of any single term and cannot be read from
873 them. The same collocation voxel with the cell mean of the contrast, and
874 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
875 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
876 second term, and with it removed the collocation and mean-only voxels differ
877 only by their constants, as in Table 3. The cell mean, with every overlapping
878 cell kept, represents the same body on every grid, and the order the layer
879 predicts is then the one observed.

880 The first-moment voxel of §8, extended to three dimensions with a con-
881 trast linear in each cell, converges at fourth order on this same sphere [4]: its
882 apparent orders are 4.01 and 4.13 where the discretisation error is isolated
883 at weak contrast, and at the full contrast its error with 16 voxels across is

884 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below the mean-only voxel's on
885 the same grid.

Table 9: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{Sa} = 0.5$		$k_{Sa} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

886 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
887 and its voxelisation changes with every grid. Table 10 shows its error for
888 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_{Sa} = 0.5$.
889 The error tracks the staircase's volume, which is 5%, 17% and 4% too large
890 on these grids. Rescaled by the ratio of the sphere's volume to the staircase's,
891 it falls below 1% at $k_{Sa} = 0.5$ and to at most 3% at $k_{Sa} = 1.0$, but it still
892 does not converge: what remains is the staircase's shape. A refinement study
893 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
894 measures the scheme.

Table 10: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

895 11. Discussion

896 Figure 5 collects the convergence of every test, with the order fitted to
897 each by least squares (`scripts/plot_convergence_orders.py`).

898 *What each basis is for.* A voxel holds two functions, and the results give
899 each basis its part. The basis of the wavefield sets how well the voxel carries
900 the field across itself: in a uniform medium the scheme converges at order
901 $2p + 2$ with the constant c_p of Eq. (34), for P and S waves and at normal and
902 oblique incidence alike. The basis of the medium sets how well the voxel *is*
903 the medium: with degree r the order is capped at $2r + 2$ (Eq. (37)). Neither
904 can make up for the other, and the second-order term shows why. The field
905 scattered once is a copy of the medium’s profile, so the wavefield basis must
906 carry the profile as well as the wave, and the error of that term is the relative
907 projection error of the profile at the lower of the two degrees (Eq. (39)). At
908 the frequencies of interest the wave varies little across a cell and the medium
909 may vary much, so it is the medium, not the wavelength, that decides the

910 degree and the size of the cells. Two cases bracket the choice. Where the
 911 medium is piecewise constant on the cells, as in a layered model with voxel
 912 faces on its interfaces, a constant is exact for the medium, the projection
 913 error vanishes, and the degree of the field is spent entirely on the wave.
 914 Where the medium varies smoothly within the cells, the two degrees should
 915 be equal: a higher degree for the contrast than for the field is wasted, and a
 916 higher degree for the field improves only the Born term. That the two degrees
 917 can be chosen independently is the practice of higher-order electromagnetic
 918 volume methods [1]; what the present results add is the rule for choosing
 919 them, and the error each choice leaves.

920 *Where the error of a voxel scheme lives.* The results separate the two ap-
 921 proximations a voxel scheme makes. The *coupling* between voxels makes
 922 none: averaged over the source cell it is the continuum's cell integral exactly,
 923 between planes by the form factor's zeros and within a plane by the tiling
 924 identity, and the self term it omits is exactly the one the single site carries.
 925 Every error therefore comes from the *single site*: from the internal field it
 926 assumes and, where the medium varies within the cell, from the medium it
 927 is given. Its T -matrix sums the rescattering within the voxel exactly, but
 928 for the voxel as represented, and the split of the second-order term in §9
 929 measures the consequence: the error sits in the part whose two scatterings
 930 lie in one cell, and the part between cells carries at most 0.2% of it. For a
 931 uniform field that error is the midpoint rule's, known in closed form; for a
 932 linear field it is the next term of the same expansion. The converse is the
 933 practical warning. A kernel that is not the cell integral (the point Green's
 934 tensor, or a cell average that does not match the single site's closure) leaves

935 a local term that does not vanish as the voxels shrink (§4): the evanescent
 936 lattice orders carry a static contribution of fixed size per voxel volume. A
 937 scheme built that way converges to the wrong limit, and refinement cannot
 938 reveal it, because the error is scale-invariant.

939 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 940 single site from a modelling choice into a derived object with a stated error.
 941 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 942 only by the cubic split of Eq. (26). Its dynamic part departs from exact
 943 scattering by a sphere only through the real $(ka)^2$ term of Eq. (27); the
 944 radiation damping is exact. That term and the collocation error of Eq. (30)
 945 have the same origin, the variation of the field across the scatterer, and the
 946 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 947 field a voxel must represent is the smooth continuum field, not the field of
 948 an isolated cube, with its concentrations at the edges and corners.

949 *Relation to the coupled-dipole method.* The same partition appears in the
 950 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
 951 relation, the lattice dispersion relation [11], the polarisability taken from the
 952 exact static solution [12] and the integrated Green tensor [13] all adjust
 953 the single site so that the lattice reproduces the continuum. For space-
 954 filling cubes the adjustment is neither needed nor a free parameter: the tiling
 955 identity fixes the local field exactly, the self term of the collocation closure is
 956 the one that makes it cancel, and what is left is a closed-form error rather than
 957 an empirical one. That the method is a collocation of the volume integral
 958 equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (29)
 959 is the same statement for the elastic lattice with the self term evaluated over

960 the finite cube, and it is what makes the error a midpoint-rule error. The
 961 filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-
 962 limit the source to the grid; the source-cell average does the complementary
 963 thing exactly, its form factor vanishing on the reciprocal lattice, and the
 964 spectral construction of Eq. (35) is the periodic coupled-dipole method of
 965 Chaumet et al. [16] with the Galerkin double average in place of the point
 966 dipole. The second-order convergence reported for the DDA by Yurkin et al.
 967 [3] corresponds to the uniform-field voxel's, and the opposite-signed constants
 968 of Eq. (30) explain why a polarisability correction can improve one observable
 969 only at the expense of another, as the package's own form-factor corrections
 970 do in reflection and transmission. Going beyond second order requires a richer
 971 internal field, which is the point of higher-order bases for volume integral
 972 equations [48, 2]; the first-moment voxel is the lowest such basis, and in its
 973 spectral form the Galerkin double average makes the lattice sum converge
 974 without Ewald splitting.

975 *The tension between the two error sources.* The coupled-dipole method has
 976 two sources of error, the single site and the coupling between sites, and each
 977 has been corrected on its own: the polarisability by the Clausius–Mossotti
 978 relation, the lattice dispersion relation or the exact static limit, and the
 979 coupling by integrating or filtering the Green tensor. The two are not in-
 980 dependent. The point kernel's evanescent orders carry a static local term
 981 whose size per voxel volume does not shrink with the voxels (§4), and the
 982 single site must cancel exactly that term. A polarisability tuned for one
 983 kernel is therefore wrong for another: the lattice dispersion relation corrects
 984 the point-dipole lattice, and changing the kernel to an integrated or a fil-

985 tered one changes the term it must cancel. Tuning either alone moves the
 986 error between the two, and refinement cannot reveal a mismatch, because
 987 the surviving term is scale-invariant. For space-filling cubes the tension is
 988 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 989 term that the closure carries (Eq. (29)), so the coupling error vanishes iden-
 990 tically and every remaining error is the single site's, the variation of the field
 991 across the cell. That error has a second tension of its own. It is a midpoint-
 992 rule error whose constants have opposite signs in reflection and transmission
 993 (Eq. (30)), so any scalar adjustment of the single site that improves one wors-
 994 ens the other; the package's far-field corrections reduce the reflection error
 995 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 996 however chosen, can remove it. What removes it is a richer internal field:
 997 the first moment lowers the error by two orders in kd in both observables at
 998 once.

999 *The cost of fourth order.* A first-moment voxel carries four times as many
 1000 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 1001 forms show what that buys. To reach an error ε in reflection, the uniform
 1002 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 1003 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 1004 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 1005 number of unknowns by about 5×10^4 (these are the Born-order constants;
 1006 at the contrast studied the first-moment voxel's measured errors lie within
 1007 5% of them, while the uniform voxel's transmission constant grows with the
 1008 contrast, so the comparison is if anything conservative). The stratified model
 1009 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-

1010 moment voxels for the same accuracy. Higher moments continue the trend
 1011 in one dimension: by Eq. (34) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 1012 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 1013 a quarter of a wavelength. In three dimensions, with the Legendre products
 1014 of total degree at most p , a voxel carries 10 ($p = 2$) or 20 ($p = 3$) weights per
 1015 state component, against 4 for the first-moment voxel. The voxel of degree
 1016 2 converges at sixth order in the layer at oblique incidence (Table 6); that of
 1017 degree 3 has not been built in three dimensions. The gain is largest precisely
 1018 where it matters: when the model varies on scales that are coarse compared
 1019 with the accuracy the wavefield must have.

1020 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 1021 mogeneous layer and a randomly stratified one, for which the discrete system
 1022 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 1023 moment voxel in a full three-dimensional solver, where the gradient channels
 1024 couple through the evanescent lattice orders (§8) and the tiling identity holds
 1025 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 1026 voxel converging at second order on a laterally heterogeneous body against
 1027 an exact solution, and the companion paper [4] shows the first-moment voxel
 1028 converging at fourth order there. A contrast that varies within a cell, which
 1029 §9 shows is needed beyond second order in a smoothly varying medium, has
 1030 a further consequence in three dimensions: its linear part is odd about the
 1031 cell's centre, so it couples the single site's uniform-field and first-moment
 1032 blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded
 1033 voxel therefore needs at least the first-moment block; that voxel is built,
 1034 and tested on the graded sphere, in the companion paper. The numerical

1035 evidence is at moderate, physically typical contrasts (up to 15% in density
 1036 and moduli) and at $ksh \leq 0.3$, the range in which the single-site T -matrix is
 1037 validated; the closed forms are first order in the contrast, and the departures
 1038 from them at the full contrast are of second order. What is established is the
 1039 *order* of convergence and its leading constant, not the behaviour at strong
 1040 contrast, where the single site's own closure may limit the accuracy before
 1041 the lattice coupling does. The single site's dynamic error is in closed form
 1042 for a sphere only; for a cube, for which no exact solution exists, it is known
 1043 through the layer (§§7–8). Two extensions follow directly. The first is the
 1044 three-dimensional solver itself. The second is a stratified *background*, a lay-
 1045 ered reference medium, as distinct from the stratified contrast of §9. There
 1046 the lattice coupling then carries the reverberation of the layering, which is
 1047 not translation invariant in depth, and whether a tiling identity holds for it
 1048 is the natural next question.

1049 12. Conclusions

1050 A voxel of a volume-integral scheme must represent the medium within it
 1051 and the wavefield within it. This paper has determined what each basis must
 1052 be and what error a given choice leaves, by an experiment in which nothing
 1053 else is approximated: a layer of cubes is the layer itself, so its exact solution
 1054 isolates the discretisation error.

1055 With the wavefield expanded to Legendre degree p and the contrast of the
 1056 medium to degree r in each voxel, the scheme converges at order $\min(2p +$
 1057 $2, 2r + 2)$. In a uniform layer the leading error is $c_p(kd)^{2p+2}$ with the closed-
 1058 form constant of Eq. (34), and a single plane of such voxels reproduces the

1059 thin-layer series term by term through $O(D^{2p+2})$; the orders 2, 4 and 6 of
 1060 the degrees 0, 1 and 2 hold for P, SV and SH waves at normal and oblique
 1061 incidence. The first-moment voxel's leading error, $-(kd)^4/720$, does not
 1062 depend on the direction of observation, whereas the collocation scheme's is
 1063 opposite in sign in reflection and transmission, so that no scalar correction of
 1064 the single site removes it. Where the medium varies within a cell, a contrast
 1065 held constant caps every voxel at second order, the coupling for a contrast
 1066 of degree r needs nothing beyond the same kernel moments to degree $p + r$,
 1067 and the error of the term of second order in the contrast is the relative
 1068 projection error of the medium at degree $\min(p, r)$, exactly in the long-wave
 1069 limit. That error is the single site's: its T -matrix sums the rescattering
 1070 within a voxel exactly for the voxel as represented, and the represented voxel
 1071 is not the medium. The two bases must therefore be matched. Where the
 1072 medium is piecewise constant on the cells the projection error vanishes: the
 1073 orders survive the interfaces of a randomly stratified model, provided voxel
 1074 boundaries coincide with them, and there the first-moment voxel reaches an
 1075 accuracy of 10^{-6} with one voxel per model layer, where the uniform-field
 1076 voxel needs about forty. A sphere whose contrast falls smoothly to zero at
 1077 its surface converges at second order against its exact solution, as the layer
 1078 predicts, to 0.018% at $k_{\text{sa}} = 0.5$ with twenty-four voxels across; a sharp
 1079 sphere's error is set by its staircase, which changes with the grid.

1080 Three properties of the cube make the experiment exact. Because cubes
 1081 tile the plane, the source-cell-averaged coupling between voxels is the con-
 1082 tinuum's cell integral exactly, and at normal incidence the lattice kernel is
 1083 available in closed form with no lattice sum. The single site is derived rather

1084 than assumed: the moments of the Green's tensor over the cube, taken as
 1085 distributions, give a closed hierarchy whose first-gradient truncation is the
 1086 uniform-strain closure. That closure is exact for a sphere in the static limit,
 1087 it departs from exact scattering by a sphere only by a real $(ka)^2$ term, with
 1088 the radiation damping exact, and for a cube it differs from the sphere only
 1089 by a cubic anisotropy of the shear response. And the collocation closure
 1090 carries the same self term as the lattice kernel, which cancels, so the discrete
 1091 system is a collocation of the continuum equation, whose second-order error
 1092 is a midpoint-rule error of the internal field, known in closed form for ev-
 1093 ery discretisation. The results, and the evidence for each, are collected in
 1094 Table 11.

1095 *The error is known in advance.* For every discretisation studied here the
 1096 error is not something to be discovered by refining until the answer stops
 1097 changing. Its order and its leading constant are known before anything is
 1098 computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the
 1099 collocation voxel (Eq. (30)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the
 1100 voxels of degree 0, 1 and 2 (Eq. (34)), and the order $\min(2p + 2, 2r + 2)$
 1101 where the medium varies within a cell (Eq. (37)), with the size of the error
 1102 there given by the relative projection error of the medium (Eq. (39)), which
 1103 is computed from the model and the grid alone. That is worth more than
 1104 the order alone, for three reasons. First, the cell size and the degree of the
 1105 voxel that a required accuracy needs follow from the wavenumber and the
 1106 medium, before the computation and without a refinement study. Secondly,
 1107 it turns a refinement study from a hope into a test. A scheme of this kind can
 1108 converge smoothly to the wrong limit (§4), and smooth convergence alone

Table 11: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (21)	sum rule (22) exact; M, Q equal an independent archive, 81/81
Channels (23)–(25)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (26)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (27)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (29)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (30)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (32)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (35)	order 4, P and S, 20° and 40°

cannot reveal it; but the difference between two grids must have the known order and the known constant, and a computation whose differences do not is in error from some cause other than the discretisation. Thirdly, it shows where effort pays. The error is that of representing the field and the medium within a cell, so raising the degree of the cell does what multiplying the cells cannot: in the stratified model one first-moment voxel per layer reaches 10^{-6} where the uniform-field voxel needs about forty. Whether the error of a body with three-dimensional structure can be known in advance in the same way is the question taken up in the companion paper [4].

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Claude (Anthropic) in order to check the existing Mathematica notebooks, Python and Fortran code, to build tests to ensure their correctness, and to write the TikZ code

1131 of the diagrams. After using this tool, the author reviewed and edited the
1132 content as needed and takes full responsibility for the content of the published
1133 article.

1134 Data availability

1135 The Mathematica notebooks and Python scripts that derive and check
1136 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)
1137 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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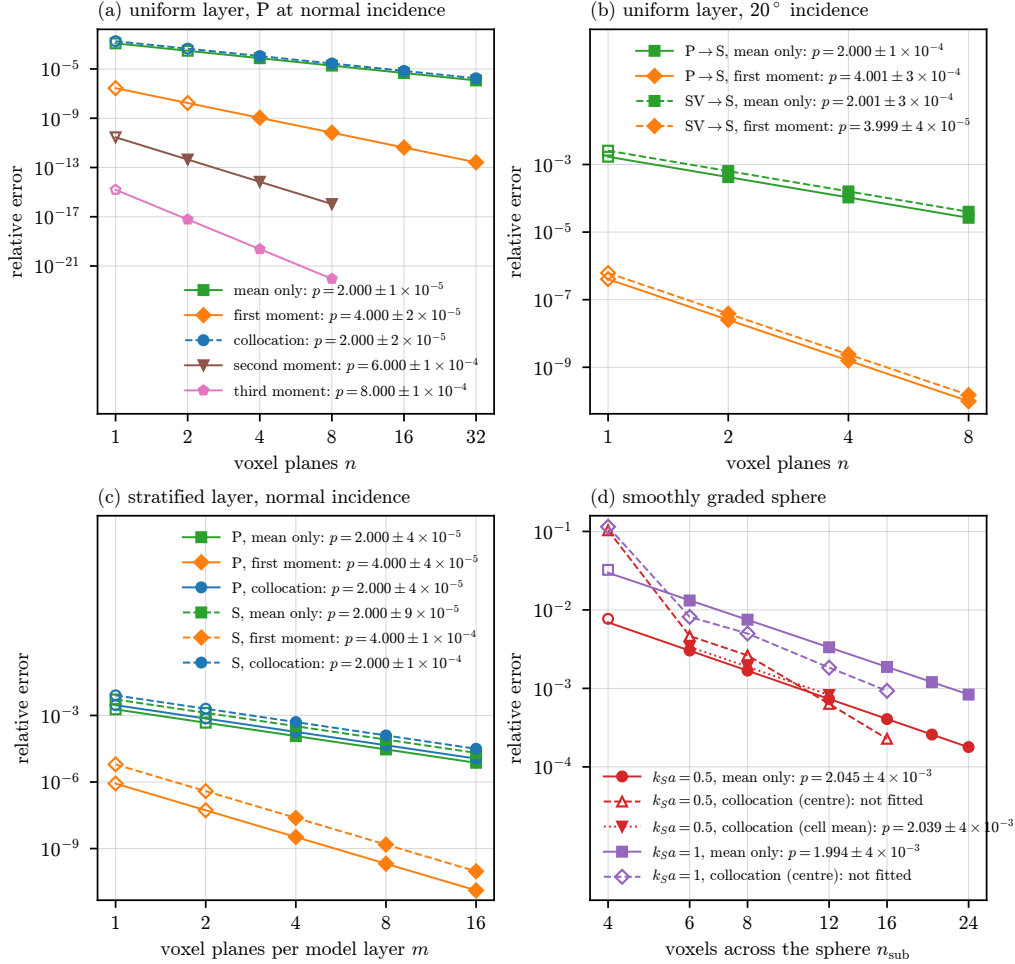


Figure 5: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree $p = 1$ (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (32) and (35)). (a) The uniform layer, P reflection at normal incidence ($D = 2\text{ m}$, $\omega = 300\text{ rad s}^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor $3/2$ of Table 3; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and